

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

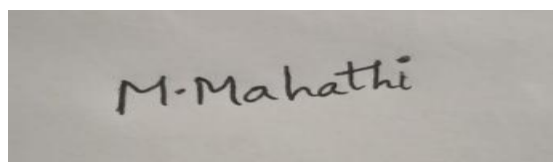
Technical Presentation

<i>Criteria</i>	<i>Technical Content Accuracy(3)</i>	<i>Problem Identification(2.5)</i>	<i>Use of Tools(2)</i>	<i>Communication(1.5)</i>	<i>Professionalism(1)</i>	<i>Total(10)</i>
<i>Weightage</i>	30%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						

Code of Conduct:

*I _M.Mahathi(192324098)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



M. Mahathi

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Topic 1: Decision Support Systems (DSS)

Focus Areas:

- Definition and need of DSS
 - Components (Data, Model, UI)
 - Types of DSS
 - Real-world applications (banking, healthcare)
 - Advantages & limitations
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Topic 2: Operational Systems vs DSS

Focus Areas:

1. OLTP vs OLAP concepts
 2. Key differences (data, users, purpose)
 3. Performance comparison
 4. Use-case scenarios
 5. Integration between systems
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Topic 3: Architecture of Data Warehouse

Focus Areas:

1. Data warehouse definition
 2. Three-tier architecture
 3. ETL process
 4. Data marts & metadata
 5. Benefits in decision-making
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Topic 4: OLAP and Data Cubes

Focus Areas:

1. OLAP concept

2. Data cube structure
 3. OLAP operations (slice, dice, roll-up, drill-down)
 4. Applications in analytics
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Topic 5: Dimensional Data Modeling

Focus Areas:

1. Fact and dimension tables
 2. Star schema vs snowflake schema
 3. Schema design process
 4. Real-world example
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Topic 6: Query Processing & BI Tools

Focus Areas:

1. Query processing stages
 2. Query optimization
 3. BI tools overview
 4. Reporting and visualization
 5. Examples of open-source BI tools
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Topic 7: OLAP Application using KNIME

Focus Areas:

1. Introduction to KNIME
 2. Workflow components (nodes, connections)
 3. Performing OLAP operations
 4. Sample application/demo
 5. Report generation
-

PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
 2. Each member must participate
 3. Assign roles:
 1. Presenter
 2. Designer (PPT)
 3. Researcher
 4. Demo/Case handler
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◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
 1. Definition and importance
 2. Core Concepts
 1. Explanation with diagrams
 3. Process/Architecture
 1. Step-by-step explanation
 4. Real-World Example
 1. Case study or application
 5. Tool/Demo (if applicable)
 1. KNIME or BI tool
 6. Conclusion
 1. Summary + future scope
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◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
 1. Diagrams
 2. Flowcharts
 3. Tables (for comparison)
3. Avoid:
 1. Too much text
 2. Copy-paste content

◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
 1. Eye contact
 2. Clear voice
 3. Logical flow



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES



CSA1603 - Data Warehousing and Data Mining

Operational Systems vs Decision Support Systems

Guided By
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Presented by:
M.Mahathi(192324098)

INTRODUCTION



Imagine Running a Supermarket

- Every time a customer buys a product, the billing system records the sale.
- At the end of the month, the manager checks which products sold the most and plans future stock.
- **Are both systems doing the same job? No!**
- One performs daily operations — the other helps in making better business decisions.



Let's understand these two important systems.



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OPERATIONAL SYSTEM



An Operational System is a computer system that handles the day-to-day activities of an organization.

Key Idea

It keeps the business running smoothly every day.
Operational System = Works Today

It performs:

- Billing
- Attendance
- Payroll
- Inventory updates
- Online transactions

Real-Life Examples

ATM Transactions

Hospital Registration

College Attendance System

Railway Ticket Booking



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Decision Support System (DSS)



A Decision Support System helps managers analyze information and make better business decisions.

Key Idea

DSS doesn't replace managers — it helps them make smarter decisions.

Examples

- Sales Analysis Dashboard
- Business Intelligence Reports
- Hospital Disease Prediction
- Weather Forecast Analysis

DSS = Plans for Tomorrow





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Operational System vs DSS



Feature	Operational System	Decision Support System
Purpose	Perform daily tasks	Help make decisions
Users	Employees, Cashiers	Managers, Executives
Data	Current transactions	Historical + Current Data
Speed	Very Fast	Analytical
Output	Bills, Receipts	Reports, Graphs
Focus	Efficiency	Improvement



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Real-Life Example



MORNING

Customer buys: Rice, Milk, Soap

Operational System:

- Generates bill
- Updates stock
- Accepts payment
- Prints receipt

Operational System works every minute.

MONTH END

Manager asks:

- Which product sold most?
- Which product is not selling?
- Which supplier gives better profit?

Decision Support System:

- Generates charts
- Predicts demand
- Suggests stock planning

DSS works when decisions are needed.



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Advantages & Limitations



Operational System

Advantages

- ✓ Fast processing
- ✓ Accurate records
- ✓ Reduces manual work
- ✓ Available 24/7

Limitations

- ✗ Cannot predict the future
- ✗ Doesn't analyze trends

Decision Support System

Advantages

- ✓ Better decisions
- ✓ Saves time
- ✓ Predicts future trend
- ✓ Reduces business risk

Limitations

- ✗ Depends on quality of data
- ✗ Requires skilled users



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Conclusion



Operational System

- Performs daily operations
- Fast and accurate
- Transaction-oriented

Decision Support System

- Helps managers
- Uses reports and analysis
- Improves decision making

*"Operational Systems keep the business running.
Decision Support Systems help the business grow."*





THANK YOU
